

The Inclusion of Preplanned and Random and Unanticipated/Unexpected Events During Strength Training Improves the Ability to Repeat High-Intensity Efforts Under Uncertainty

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Purpose: To compare the effects of unilateral flywheel training (FT), using a rotational conical pulley, including multidirectional movements with either preplanned or random unanticipated/unexpected executions on functional performance in football players. **Methods:** A total of 32 young male football players were randomly assigned to an FT program including preplanned unilateral multidirectional movements (PTG, $n = 11$), an FT executing the same unilateral movements through random (ie, right or left leg) unanticipated (ie, verbal or visual cue) or unexpected (ie, moment where the cue was provided) situations (UTG, $n = 11$), or a control group ($n = 10$) that followed their football training routine. FT consisted of 1 set $\times$ 5–12 repetitions of 4 exercises performed once a week for 10 weeks. Intermittent endurance performance, repeated unilateral and bilateral jumping ability, change-of-direction (COD) ability, linear sprint velocity, preplanned repeated-sprint ability (RSA), and uncertainty RSA (RSA-RANDOM) were assessed preintervention and postintervention. **Results:** Significant improvements were found in RSA-RANDOM performance ($P < .05$, effect size [ES] range: UTG [1.83–2.16], PTG [0.69–0.73]) and COD ($P < .05$, ES: UTG = 1.34, PTG = 0.98) in both training groups. Furthermore, significant improvements were also found in intermittent endurance performance ($P = .016$, ES = 0.37) and sprinting ($P = .006$, ES = 0.45) in UTG. No changes in any variable were found in the control group. No significant between-groups differences ($P > .05$) were reported between UTG and PTG, while differences were observed to the control group in unilateral jumping ability, COD, and RSA-RANDOM for UTG, and in RSA-RANDOM for PTG. **Conclusions:** A 10-week unilateral FT improved RSA-RANDOM and COD ability in youth football players, so both preplanned and unexpected situations should be included on strength training.

Keywords: resistance training, repeated-sprint ability, variable training, preactivation, feedforward

In elite football, the final aim of a strength training program should be to optimize how players effectively move on the pitch and maintain high-intensity efforts over the match/season while reducing the injury rate (ie, limiting layoff time from competition and practice). However, limited time is available for strength and conditioning during the in-season period, given to highly congested calendars¹ and, consequently, increasing the injury rate.² This reality challenged us to seek for more time-efficient training strategies to maintain or even create the adaptations needed to tailor those aims.^{3,4}

As Tous-Fajardo³ suggests, football players perform most of their specific actions in an unexpected (ie, unpredicted); highly dynamic; and stochastic environment that leads players to adopt anticipatory behaviors (feedforward mechanisms and co-contractions)⁵ or generate new movement strategies. Specifically, the nature of most football actions is unilateral, variable, and multidirectional,⁶ such as kicking, passing, pressing, cutting, or jumping while mainly using 1 limb over the 3 space dimensions. Rather than

repeating these moves cyclically, they alternate them without a prefixed route or pacing strategy resulting in ad hoc duration. Several perturbations,⁷ unknown,⁸ or unexpected⁵ events coming from the opponents or teammates usually constrain their preplanned movement strategies, leading to a sudden breaking action producing high amounts of eccentric force,⁴ which is often required to reorganize the plan or fake the defender. Consequently, there is a need to introduce more challenging training methods, including the previously mentioned variables that allow players to perceive and react to the environment that may induce emergent behaviors for generating optimal movement synergies.⁹

During the past few years, the effects of different training programs, including some of these variables, have been evaluated. Thus, several force vector-specific flywheel training (FT) programs have found beneficial effects in various abilities such as changing of direction (COD),^{4,6,10} linear sprinting,^{4,6,10} jumping,^{6,11} throwing,¹² or high-intensity repeated efforts.¹² Specifically in footballers, a combination of flywheel multidirectional movements (eg, lateral squat, backward lunges, diagonal trunk rotations, and unilateral hamstrings “kicks”) and unilateral squats with superimposed vibration found COD and linear sprinting improvements compared to a conventional combined strength training program.⁴ Furthermore, a group of young footballers achieved substantial enhancements in unilateral jumping after a unilateral lateral squat performed through FT.¹¹ In addition, significant improvements in

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linear sprinting, COD, and shooting were reported in young footballers after a combination of FT with plyometrics in comparison to a football training group.¹⁰ In this regard, few studies^{10,12} have superimposed random unexpected constraints on functional movements. However, despite obtaining beneficial effects in a wide range of athletic performance tests (ie, preplanned tests), the effects of including random and unexpected stimuli during FT remain unknown in unexpected tests (ie, a new form of assessment).

To the best of our knowledge, studies have yet to compare the effects of training programs, including accentuated eccentric contractions and random; unexpected; and unanticipated (ie, the possibility of imperative cues being of different nature) actions in male football players on specific preplanned and unexpected tests. Therefore, this study aimed to compare the effects of a unilateral FT, including multidirectional movements with either preplanned or random, unexpected, and unanticipated executions, on a wide battery of functional preplanned and unexpected tests. Our hypothesis is that all training groups will exhibit improvement in both preplanned and unexpected tests. We anticipate that the group exposed to random, unexpected, and unanticipated conditions will achieve superior results, particularly during unexpected tests.

Methods

Subjects

A total of 32 male young football players (15.5 [0.8] y, 174.6 [67] cm, and 62.5 [14.1] kg) belonging to the same semiprofessional football club voluntarily participated in this study. Sample size was estimated a priori using G*Power (G*Power 3.1.9.2, Heinrich Heine-Universitat Dusseldorf; <https://www.gpower.hhu.de/>). The effect size (ES) was computed using the means and between-subject standard deviations (SDs) from a previously published study¹¹ that investigated the effects of unilateral FT, including multidirectional movements on vertical jump. The means and SDs were 33.9 (3.8) and 36.0 (4.5) cm of counter-movement jump height, respectively. Resulting in a Cohen *d* ES of 0.48, which can be classified as and so equivalent to an *f*=0.3 (moderate). The average SD was used to compute the ES. Alpha was set to 5%, while power was set at 80% (1 – β). The estimated total sample size for and ES *f* of 0.25 was 24 participants (actual power=0.808), but considering possible dropouts, we enrolled

32 participants in this study that were randomly allocated to one of the 3 conditions. All players participated, on average, in 4 football training sessions (120 min each) and 1 official match per week (Table 1). Players had at least 1 year of resistance training experience. They had no history of regular FT and no lower limb muscle or bone injury in the past 2 months. Participants were excluded from the study when they did not attend at least 80% of training sessions (ie, regular football sessions and prescribed training program); when they did not complete all the assessment sessions or when any given injury occurred during a scheduled practice or competition that caused absence from the next training session or match. Three players in the control group (CON) and 1 in the preplanned training group (PTG) were excluded for various reasons (ie, injury [n=2], illness [n=1], or lack of interest [n=1]). None of the players were injured during the eccentric overload training sessions. The final sample sizes for the training groups were 11 for unexpected training group (UTG), 10 for PTG, and 7 for CON. Despite dropouts, no significant differences were found between groups at baseline. The final sample had the following characteristics: 15.6 (0.8) years, 173.9 (67.1) cm, and 62.6 (14.0) kg. Players and parents/guards were informed of the study’s procedures, potential risks/benefits, and both signed a written informed consent before starting the investigation. The study was conducted according to the Declaration of Helsinki (2013), and the protocol was fully approved by the ethics committee of ***for blinding purpose*** before recruitment. Participants were asked not to perform strenuous exercise the day before each testing session and to consume their last meal (caffeine-free) at least 3 hours before the scheduled testing time.

Design

Using a controlled and randomized study design, subjects were divided into an UTG (n=11), a PTG (n=11), or a CON (n=10; <https://www.randomizer.org/>). Three weeks before starting the intervention, subjects were familiarized with testing procedures and exercises. During the next 10 weeks, they performed the experimental training protocol once per week (day –3 within the microcycle) in addition to their regular football training. One week before and after the intervention intermittent endurance performance, repeated unilateral and bilateral jumping ability, COD ability (*T* test), linear sprint velocity, preplanned repeated

Table 1 An In-Season Weekly Program for the Football Players

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Recovery (60 min)	Rest	Strength (90 min)	Endurance (90 min)	Speed (80 min)	Rest	Match
Warm-up: 15 min		FT (preplanned <i>and non-pre-planned movements group</i>)	Warm-up: 15 min	Warm-up: 15 min		
Starters		Warm-up: 15 min	Positional games: 20 min	Medium-sided game: 25 min		
Aerobic endurance: 15 min		Positional games: 20 min	Large-sided game: 20 min	Positional games: 15 min		
Injury prevention: 30 min		Small-sided games: 25 min	Simulated game: 35 min	Tactical drills: 15 min		
Nonstarters		Tactical drills: 30 min		Strategy: 10 min		
Small-sided games: 30 min						
Speed endurance: 15 min						
Injury prevention: 15 min						

Abbreviation: FT, functional flywheel training 20–30 min plus regular football training 90 min.

sprint ability (RSA), and uncertainty RSA (RSA-RANDOM) were assessed.

Methodology

Training Intervention

Participants in both training groups (UTG and PTG) performed 1 weekly additional training session,¹³ always in the afternoon (6–8 PM) for 10 weeks, involving a total of 20/30 minutes per session. CON continued with their regular football training (please see Table 1), and they did not perform any strength training. Both UTG and PTG consisted of 1 set of 4 unilateral exercises: front step, backward lunges, defensive-like shuffling steps, and lateral crossover cutting (Figure 1) using a portable conical pulley (VersaPulley, Costa Mesa; inertial 0.27 kg/m², transmission pulleys/harness setup as shown in Figure 1). While PTG first executed all repetitions with the nondominant limb and right after that, with the dominant one, UTG randomly received either a verbal (ie, the trainer said right or left) or visual (ie, move to the same side as the trainer or where the ball was) cue to perform each repetition with the corresponding limb (ie, right or left; please, see [Supplementary Material S1 to S4](#) [available online]). There was a random (ie, right or left); unanticipated (ie, verbal or visual cue); and unexpected (ie, moment when the cue was provided) execution in the UTG. To balance each set, each leg and each cue (verbal or visual) performed half the total number of repetitions. (This information was blinded for players.) The dominant leg was defined as the preferred kicking leg. A progressive volume-based overload was developed (5–12 repetitions, and speed:force ratio of 1–2 out of 4, for details, please see Table 2). All sessions were

supervised by the strength and conditioning staff of the team, providing adequate feedback. Participants were encouraged to perform the concentric phase as fast as possible while delaying the braking action to the last third of the eccentric phase, following the recommendations for FT.^{3,4,6}

Functional Performance Tests

Functional performance tests were performed on 3 nonconsecutive days (ie, 48 to 72 h between testing days) before training intervention, with a 40-m linear sprint test and 30 to 15 intermittent fitness test administered during the first day, jumping tests and RSA test executed during the second day, and finally, the COD test and RSA-RANDOM test completed on the third day. All tests were performed at the same time (6–8 PM, temperature 17–22 °C, and humidity 60%–70%) on the regular grass-football field wearing football-specific boots.

Intermittent Endurance Performance

The 30-15 Intermittent Fitness Test was evaluated as described elsewhere.¹⁴ Velocity (in kilometers per hour) attached in the last completed stage (V_{IFT}) was considered for further analysis.

40-m Linear Sprint Test

Running speed was evaluated by the 40-m sprint test. Time was registered with a double-beam time gate system (Witty, Micro-gate), which was placed at 0.75 m height and 1.5 m distance between each other. Participants were requested to place the front foot 0.5 m before the first timing gate while adopting a 2-point staggered stance. Two repetitions separated by at least 2 minutes of

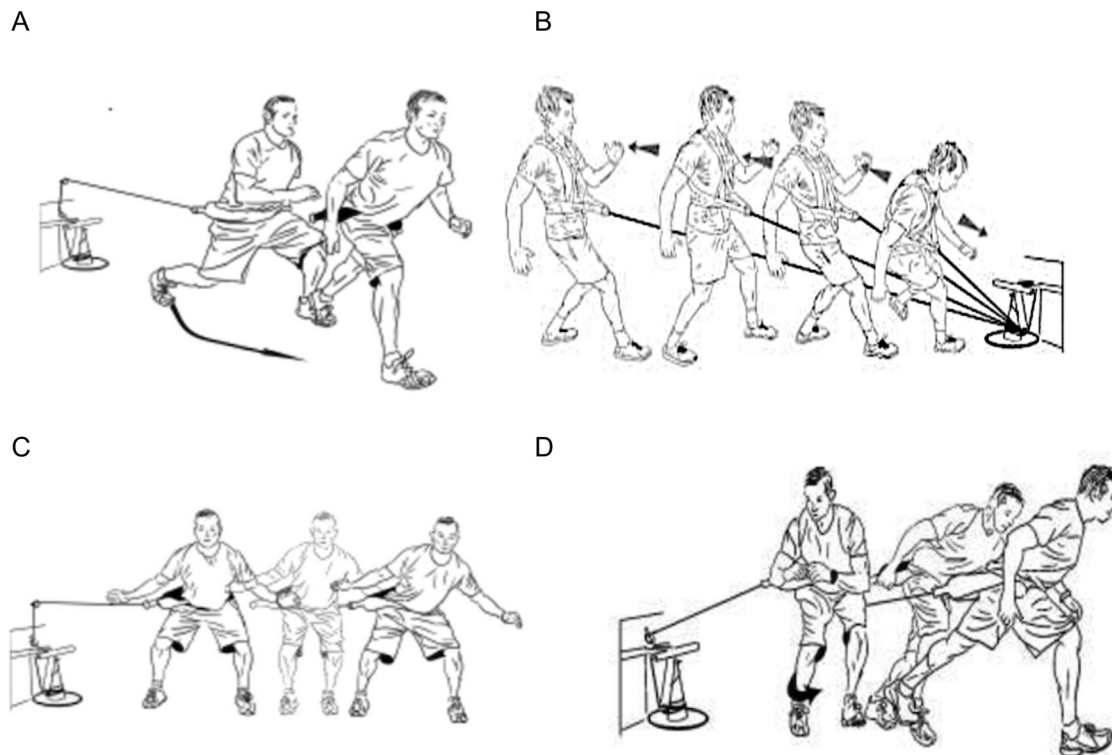


Figure 1 — Functional eccentric overload unilateral training program: (A) front step, (B) backward lunges, (C) defensive-like shuffling steps, and (D) lateral crossover cutting.

Table 2 Functional Flywheel Training Program With Preplanned and Unexpected Movements

Week	Exercise	Intensity ^a	Sets × repetitions	Recovery ^b
1	(1) Front step	6–1	1 × 5	2 min
2	(2) Backward lunge	6–1	1 × 5	
3	(3) Defensive-like shuffling steps	6–1	1 × 6	
4	(4) Lateral crossover cutting	6–1	1 × 6	
5		6–1	1 × 8	
6		6–2	1 × 8	
7		6–2	1 × 10	
8		6–2	1 × 10	
9		6–2	1 × 12	
10		6–2	1 × 12	

^a6–1 denotes 6 loads (0.27 kg/m²) with the first height level in conical pulley. ^bRecovery between exercises.

passive recovery were performed. The best time was used for further analysis.

Triple-Hop Test

Triple-hop-for-distance test with dominant (3HopD) and nondominant (3HopND) leg and bilaterally was performed.¹⁵ Two trials of each test interplead by 1 minute of passive recovery were completed. Hop distance was measured in cm from the toe at the push-off to the heel where the subject landed. Only the best attempt was used for further analysis.

COD Ability

The modified *T* test was used to assess COD ability,¹⁶ using the same setup and system described for 40-m test. Two trials, with 30-second recovery, were allowed, and the best result was included in the data analysis.

RSA Test

The RSA test involved 6 repetitions of maximal 2 × 15-m shuttle sprints with 20 seconds of recovery.¹⁷ During the recovery, players were required to stand passively. Six seconds before starting each sprint, the participants were asked to assume the start position and await the start signal through a 3-second countdown provided by a light panel (Microgate). Three scores were calculated for the RSA test: the best sprint time, the slowest sprint time, and the mean sprint time. The same beam, time gate system previously described was used for RSA.

RSA-RANDOM Test

The test consisted of 6 × 20-m (10-m + 10-m) sprints separated by 20 seconds of active recovery while jogging back to the initial position.¹⁸ Each repetition involved a 10-m linear sprint plus a 10-m sprint with 3 potential directional options according to a traffic light fixed after the first 10-m linear sprint: (1) linear sprint, (2) linear sprint after performing a 45° COD to the right, and (3) linear sprint after performing a 45° COD to the left. To balance the test, all sprint directional options were repeated twice for 6 repetitions. (This information was blinded for players.) Six seconds before starting each sprint, the participants were asked to assume the start position and await the start signal through a 3-second countdown provided by a light panel (Microgate). Three scores were calculated for the RSA-RANDOM test: the best sprint time, the slowest sprint time, and the mean sprint time.

Statistical Analysis

Data are presented as mean (SD). To prove the normality of data distribution and the homogeneity of variances, the Kolmogorov–Smirnov and Levene tests were conducted. A paired-sample *t* test was used to evaluate within-group differences. Also, an analysis of covariance was performed to detect possible between-group differences, assuming baseline values as covariates. A post hoc comparison (Bonferroni correction) was used to examine mean differences among groups (ie, UTG, PTG, and CON). The statistical significance was set at $P < .05$, and the adjusted *P* value was calculated for all tests. The standardized difference or ES (95% confidence limit) in the selected variables was calculated using the pooled pretraining SD. Threshold values for Cohen ES statistics ≤ 0.20, negligible; 0.21 to 0.49, small; 0.50 to 0.79, medium; 0.80 to 1.29, large; and ≥ 1.30 very large.¹⁹ All statistical analyses were performed using SPSS (version 25) and Microsoft Excel (version 2016).

Results

In Table 3 are presented the intragroup differences after the training period. Significant improvements for UTG were found in V_{IFT} ($P = .004$) and 40-m ($P = .001$), as well as in RSA-RANDOM performance (best [$P = .001$], mean [$P = .001$], and worse [$P = .001$]) and *T* test ($P = .001$). PTG improves their performance in RSA-RANDOM test (mean [$P = .001$] and worse [$P = .020$]) and *T* test ($P = .001$). No changes in any variable were found in CON.

Between-group differences after the training period are shown in Table 4. No significant between-group differences ($P > .05$) were reported between UTG and PTG. Significant differences were observed in unilateral jumping ability (triple-hop-for-distance test with dominant leg [$F = 7.661$; $t = 3.867$; $P = .024$] and triple-hop-for-distance test with nondominant leg [$F = 8.128$; $t = 3.878$; $P = .024$]), *T* test ($F = 9.327$; $t = -4.024$; $P = .018$), and RSA-RANDOM mean ($F = 19.319$; $t = -5.993$; $P = .001$) and worse ($F = 12.327$; $t = -4.579$; $P = .001$) between UTG and CON. Also, significant differences between PTG and CON in RSA-RANDOM mean ($F = 19.319$; $t = -4.828$; $P = .001$) and worse ($F = 12.327$; $t = -4.143$; $P = .018$) were found.

Discussion

This study aimed to compare the effects of unilateral FT including multidirectional movements with either preplanned or random

Table 3 Differences After the Training Period in UTG, PTG, and CON Groups, Mean (SD)

Variables	UTG, n = 11					PTG, n = 10					CON, n = 7				
	Pre	Post	<i>P</i>	<i>P</i> -ad	ES	Pre	Post	<i>P</i>	<i>P</i> -ad	ES	Pre	Post	<i>P</i>	<i>P</i> -ad	ES
V _{IFT}	18.68 (1.21)	19.18 (1.15)	.004	0.016	0.37	19.0 (1.18)	19.25 (1.15)	.363	0.545	0.21	19.6 (0.93)	19.4 (0.99)	.407	0.488	−0.21
3HopD	6.50 (0.40)	6.69 (0.46)	.056	0.168	0.43	6.52 (0.4)	6.62 (0.39)	.240	0.411	0.18	6.16 (0.33)	5.94 (0.34)	.025	0.100	−0.60
3HopND	6.57 (0.46)	6.74 (0.4)	.059	0.142	0.33	6.58 (0.35)	6.68 (0.38)	.111	0.222	0.28	6.35 (0.32)	6.14 (0.27)	.110	0.264	−0.57
Bilateral	6.96 (0.61)	6.94 (0.58)	.818	1.091	−0.03	7.11 (0.39)	7.23 (0.35)	.075	0.180	0.29	6.57 (0.34)	6.53 (0.24)	.678	0.740	−0.07
<i>T</i> test	9.26 (0.30)	8.81 (0.29)	.000	0.001	1.34	9.20 (0.38)	8.79 (0.34)	.000	0.001	0.98	9.31 (0.15)	9.22 (0.15)	.140	0.280	0.49
40 m	5.29 (0.14)	5.22 (0.11)	.001	0.006	0.45	5.30 (0.21)	5.26 (0.13)	.064	0.192	0.15	5.56 (0.22)	5.51 (0.23)	.348	0.464	0.17
RSA-Rb	3.14 (0.08)	2.97 (0.08)	.000	0.001	1.87	3.11 (0.13)	3.01 (0.11)	.029	0.116	0.73	3.06 (0.09)	3.06 (0.10)	.010	0.060	0.04
RSA-Rm	3.27 (0.06)	3.12 (0.10)	.000	0.001	2.16	3.20 (0.13)	3.10 (0.1)	.000	0.001	0.70	3.18 (0.08)	3.22 (0.08)	.006	0.072	−0.37
RSA-Rs	3.38 (0.07)	3.24 (0.11)	.000	0.001	1.83	3.29 (0.12)	3.20 (0.08)	.002	0.012	0.69	3.29 (0.09)	3.35 (0.09)	.332	0.569	−0.59
RSA-180b	5.52 (0.13)	5.49 (0.12)	.209	0.418	0.21	5.52 (0.15)	5.54 (0.17)	.741	0.808	−0.06	5.67 (0.12)	5.74 (0.10)	.332	0.569	−0.51
RSA-180m	5.74 (0.10)	5.71 (0.09)	.317	0.543	0.27	5.70 (0.14)	5.71 (0.18)	.658	0.877	−0.06	5.86 (0.10)	5.91 (0.14)	.070	0.210	−0.45
RSA-180s	5.94 (0.09)	5.92 (0.11)	.597	0.896	0.20	5.84 (0.16)	5.83 (0.19)	.729	0.875	0.07	6.04 (0.10)	6.05 (0.16)	.864	0.864	−0.08

Abbreviations: Bilateral, distance achieved in the bilateral horizontal test; CON, control group; ES, effect size; 3HopD and 3HopND, distance reached in the horizontal triple dominant and nondominant hop test; 40-m, time in the 40-m linear sprint test; *P*, level of significance; *P*-ad, adjusted *P* value; PTG, preplanned; RSA-Rb, RSA-Rm, and RSA-Rs, best, mean, and slowest time in the repeated-sprint ability random test; RSA-180b, RSA-180m, and RSA-180s, best, mean, and slowest time in the repeated-sprint ability test with a 180° change of direction; *T* test, performance in the change-of-direction *T* test; UTG, unexpected; V_{IFT}, final velocity reached in the 30-15 Intermittent Fitness Test. Positive ES values are considered an improvement, and negative results, an impairment.

Table 4 Between-Groups Differences After the Training Period

Variables	<i>F</i>	UTG vs PTG			UTG vs CON			PTG vs CON		
		<i>t</i>	<i>P</i>	<i>P</i> -ad	<i>t</i>	<i>P</i>	<i>P</i> -ad	<i>t</i>	<i>P</i>	<i>P</i> -ad
V _{IFT}	1.362	0.658	1.000	1.636	1.649	.337	0.808	1.07	.864	1.481
3HopD	7.661	0.986	1.000	1.636	3.867	.002	0.024	2.946	.021	0.151
3HopND	8.128	0.621	1.000	1.636	3.878	.002	0.024	3.267	.010	0.090
Bilateral	2.777	−1.757	.275	0.707	0.838	.100	0.400	2.218	.109	0.392
<i>T</i> test	9.327	−0.277	1.000	1.636	−4.024	.001	0.018	−3.672	.04	0.180
40-m	1.290	−1.173	.757	1.434	−1.456	.475	1.005	−0.528	1.000	1.636
RSA-Rb	4.214	−1.233	.689	1.378	−2.903	.023	0.138	−1844	.233	0.699
RSA-Rm	19.319	−1.554	.399	0.897	−5.993	.000	0.001	−4.828	.000	0.001
RSA-Rs	12.327	−0.693	1.000	1.636	−4.579	.000	0.001	−4.143	.001	0.018
RSA-180b	1.625	−0.925	1.000	1.636	−1.766	.270	0.747	−0.901	1.000	1.636
RSA-180 m	4.151	−1.116	.766	1.378	−2.881	.025	0.128	−1.878	.218	0.713
RSA-180s	0.323	0.109	1.000	1.636	−0.736	1.000	1.636	−0.722	1.000	1.636

Abbreviations: Bilateral, distance achieved in the bilateral horizontal test; CON, control group; 3HopD and 3HopND, distance reached in the horizontal triple dominant and nondominant hop test; 40-m, time in the 40-m linear sprint test; *P*, level of significance; *P*-ad, adjusted *P* value; PTG, preplanned group; RSA-Rb, RSA-Rm, and RSA-Rs, best, mean, and slowest time in the repeated sprint ability random test; RSA-180b, RSA-180 m, and RSA-180s, best, mean, and slowest time in the repeated sprint ability test with a 180° change of direction; *T* test, performance in the change of direction *t* test; UTG, unexpected group; V_{IFT}, final velocity reached in the 30 to 15 intermittent fitness test.

unexpected executions, on a wide battery of functional preplanned and unexpected tests. The main finding was both training groups reported significant differences in RSA-RANDOM and COD ability.

This is the first study that analyzes the ability to repeat sprints with CODs in random unexpected order. In this regard, although no between-group significant differences were found, UTG showed greater performance than PTG ($ES_{\text{within}} = 2.16$ vs 0.70 ; $ES_{\text{between}} = 0.52$). In addition, significant differences between both training groups compared to CON in RSA-RANDOM mean and worse were obtained. As no similar studies have been previously developed, comparisons are not possible. However, the inclusion of unknown loads during strength training has been shown to increase muscle preactivation and subsequent force production (ie, rate of force development)²⁰ and the stretch-shortening cycle performance,²¹ which in turn leads to sport-specific improvements (eg, handball-throwing velocity).⁸ Furthermore, tripping or bumpy motion during executions may have led to more muscle activation²² and intensity due to the postural adjustments.²³ It is interesting to note that including some physical perturbations (eg, pushing the athletes in different random directions), which directly affect athletes' motion during execution, may lead to similar results. However, one of the possible considerations to argue between-group differences is a matter of specificity. The UTG group trained during the training program in reaction time tasks, whereas the PTG group did not. Thus, task specificity might be another argument to support the current findings. In this regard, the UTG included 3 different constraints by responding to a random (right or left), unanticipated (verbal or visual), and unexpected (moment when the cue was provided) stimuli which may induce not only a more proactive (as opposed to reactive) behavior but also an increase in early force production to optimize physical performance, as it occurred in this study with RSA-RANDOM. Therefore, the inclusion of unanticipated executions during neuromuscular training might be a key factor in improving repeated high-intensity efforts involving multiple-choice situations.

Regarding COD ability, our results revealed significant improvements ($P < .05$) in both training groups (UTG, $ES = 0.98$; and

PTG, $ES = 1.34$), which are within the range of or even greater than those of previous studies ($ES = 0.04$ – 2.73) including functional FT (ie, side step, lateral step, or lateral crossover step performed with a conical pulley).^{4,6,10} On the other hand, substantial improvements have been found in COD speed,^{4,6,10} irrespective of the COD test used, but with a wide variety of results ($ES = 0.22$ – 2.73). The player's age experience (adults vs late adolescents), training load distribution (biweekly vs weekly), number of exercises, and test differences in both the number of turns (one vs several) and the distances covered can explain the between-study differences. Notwithstanding, the inclusion of brief episodes of more demanding eccentric actions, such as it may occur with FT, and more specific (eg, lateral or rotational) force vector application seems to be critical factors in developing specific COD adaptations.⁴

In addition, significant results were only found in the 40-m test for UTG ($ES = 0.45$, $P < .05$). These results are in accordance with those reported after different FT programs in team-sport athletes ($ES = -0.23$ to 0.95).^{4,6,24,25} It is worth noting that higher changes ($ES = 0.84$) were observed after an 8-week leg curl FT program in 30-m linear sprinting from a stop starting position.²⁵ Interestingly, while the 20-m sprint achieved possible benefits ($ES = 0.32$) after a combined FT compounded by half-squats and leg curls, the 10-m flying time showed an almost certainly effect ($ES = 0.95$). In this regard, it has been shown that the higher peak force produced by the hamstrings the faster speed was achieved,²⁶ which also correlates well with those athletes who are able to highly activate these muscles just before ground contact, producing higher eccentric peak torques,²² meaning that if longer distances are covered (ie, 40 m) more hamstring musculature involvement may be expected. Surprisingly, including random unexpected events during the execution of the selected exercises substantially increased linear sprinting performance compared to executing the same exercise but staged in a preplanned situation. It may be possible that uncertainty increased neuromuscular preactivation^{20,21} before and during sprinting, improving force production through greater efficiency. In addition, the unexpected constraints in UTG frequently produced tripping or bumpy motion with subsequent balance recovery rapid responses. Research from fall prevention

has shown that after tripping, the most pronounced early muscle responses are observed in the hamstrings and gastrocnemius muscles.²³ These responses (with latencies of 60–80 ms) are nonspecific but highly functional as they provide the hip and ankle extension moments and knee flexion moments required for successful stability recovery.²³ It may be that this potential rapid neural activation (or even preactivation) before ground contact (once subjects anticipate the loss of stability) of the hamstrings/triceps surae muscles is a plausible explanation of the linear speed increases, so it deserves further investigation. Moreover, as previously reported, including only one session per week⁴ or performing exclusive training based on squatting²⁴ has shown trivial effects, as in PTG or CON. Therefore, training programs to enhance long sprinting performance should involve eccentric hamstring exercises (ie, hip extension and knee flexion); horizontal force exercises; 2 weekly sessions; unexpected events; and the combination altogether.

To our knowledge, this was the first time that intermittent endurance performance has been analyzed after a strength training program. Although no significant between-group (UTG vs PTG) differences ($P > .05$) were found, UTG developed significantly higher performance. In this regard, V_{IFT} is considered a crucial factor of global performance,¹⁴ showing moderate to very large relationships with various key performance indicators (eg, RSA, 10-m, maximal oxygen uptake, and countermovement jump). As it includes 40-m running distances with COD repeated over time, previous improvements in such abilities (COD, jumping, RSA, and sprinting) may help to increase intermittent performance. However, future research is warranted to better understand the strength training influence on intermittent endurance performance.

Although no significant repeated unilateral horizontal jumping differences were found between experimental groups, UTG showed greater ESs compared to PTG (0.33–0.43 vs 0.18–0.28). As no information about the effects of strength training on repeated horizontal jumping is currently available in the literature, comparisons are not possible. However, there are several studies using single-leg horizontal hop tests finding similar (ES = 0.05–0.39)²⁷ or even greater magnitudes (ES = 0.64–0.65) after FT.²⁸ It seems that repeated unilateral jumping needs greater neuromuscular preactivation before landing and preparing for pushing off as fast as possible. However, bilateral jumping has shown greater improvement in PTG than UTG (ES = 0.97). Such controversy might be due to different mechanisms related to jumping biomechanics and, consequently, to the training program performed by each group. While unilateral jumping requires stabilizing before pushing off as fast as possible (players needed to be balanced before each unexpected repetition to apply force), bilateral jumping might involve a lower stabilizing strategy (as during the PTG) and mainly impact the ability to push-off as fast as possible. Therefore, future studies should be developed to understand the main jumping adaptation mechanisms through different strength training strategies.

Practical Applications

Football and other situational sports are stochastic in nature, and players constantly adapt to new contexts. The inclusion of different constraints through random, unanticipated, and unexpected executions may optimize the perceptual and cognitive mechanisms needed to better interpret game situations and reduce premotor time. To “predict the unpredictable” or “expect the unexpected” are useful watchwords we often repeat in our football training sessions

to spur and shape emergent functional behaviors in our football players. Finally, practitioners should consider the transfer of these concepts into their football training schedules by alternating actions such as passing or heading a ball, jumping, and turning, pushing or not against an opponent, and combining attack/defense situations in base to a specific stimulus in order to challenge central nervous system avoiding simple automatisms.

Conclusions

Flywheel training implementing sport-specific movements has been proved to improve different performance-related measurements in youth football players. Indeed, including both preplanned and random unexpected situations within each repetition seems to be a critical factor in developing the ability to repeat unanticipated high-intensity efforts.

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